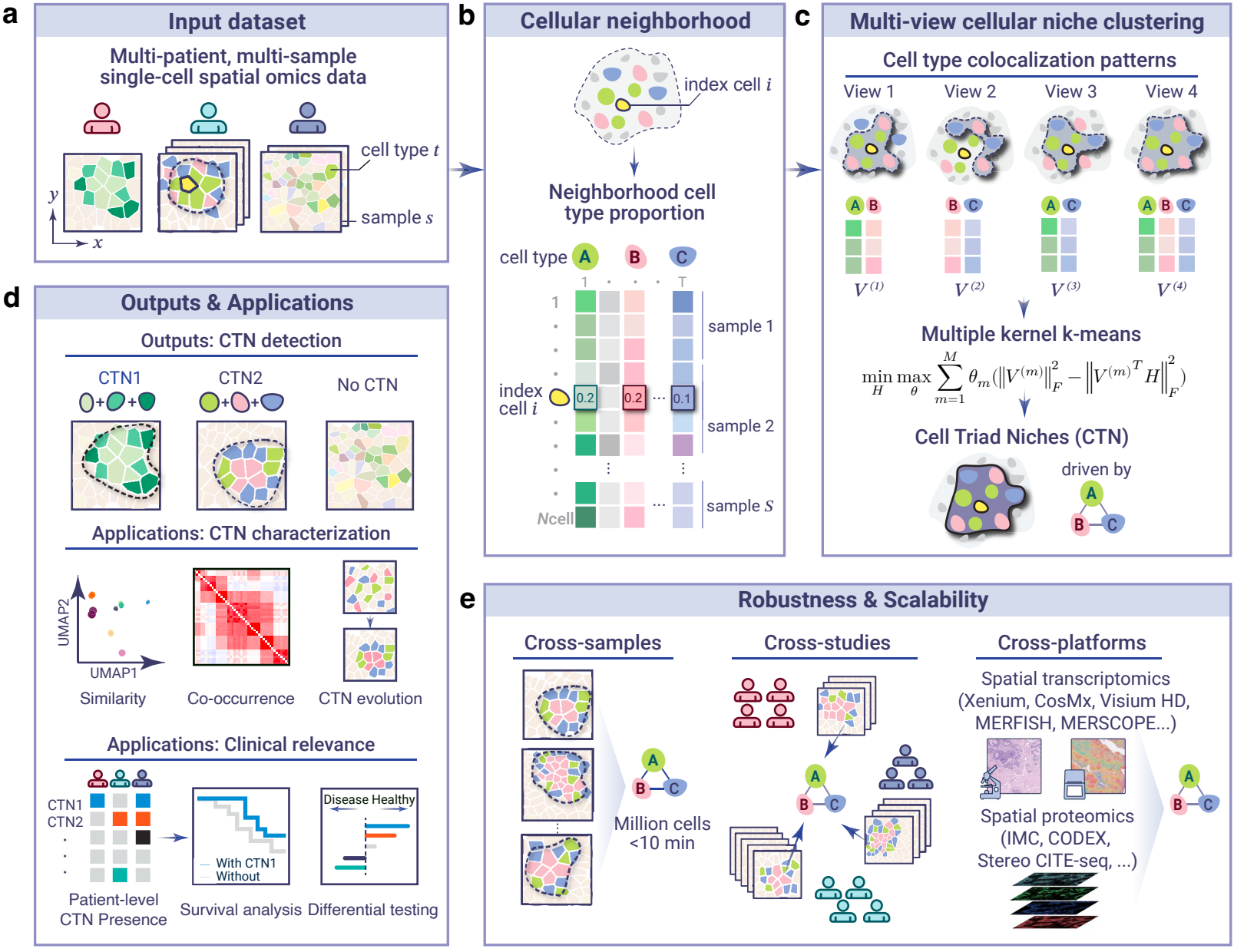

a
Input dataset
Multi-patient, multi-sample single-cell spatial omics data
cell type t
sample s
y
x
b
Cellular neighborhood
index cell i
Neighborhood cell type proportion
cell type A B C
index cell i
sample 1
sample 2
sample S
Ncell
c
Multi-view cellular niche clustering
Cell type colocalization patterns
View 1
View 2
View 3
View 4
Multiple kernel k-means
Cell Triad Niches (CTN)
driven by
d
Outputs & Applications
Outputs: CTN detection
CTN1
CTN2
No CTN
Applications: CTN characterization
UMAP2
UMAP1
Similarity
Co-occurrence
CTN evolution
Applications: Clinical relevance
CTN1
CTN2
Patient-level CTN Presence
With CTN1
Without
Survival analysis
Disease Healthy
Differential testing
e
Robustness & Scalability
Cross-samples
Million cells <10 min
Cross-studies
Cross-platforms
Spatial transcriptomics (Xenium, CosMx, Visium HD, MERFISH, MERSCOPE...)
Spatial proteomics (IMC, CODEX, Stereo CITE-seq, ...)